

Project Overview

- Define Business requirements for CTU training solutions
- Identify Azure Services Required
- Create Azure resource groups in azure for students
- Document region tags and configuration

Business Requirements for CTU

Core Requirements

Students will need to be able to access Azure from anywhere for example if the student is sick, they would need to be able to access the data on azure from their home. Labs would be required for the students in the IT field to be able to practice and develop skills within their study field

There would need to be a storage blob where course materials can be stored to help the students. The materials that would be stored are videos of the lectures, Slides and homework/ past papers. This storage would also need to be easily accessible by any student from home or campus

The accounts of each user would be managed and would fall within 3 groups. These groups are the following Students, Lecturers and Administrators. The Administrators would need to provide a secure login and access permission for all the users

The infrastructure would need to scale with the number of students each semester. This is important since if the infrastructure does not scale with the amount of students, then the performance of this cloud would decline quickly, and the cost of the cloud might go above the intended budget.

Security would need to be a top priority since a lot of sensitive information would be stored, such as student information and academic records. Implementing RBAC would increase the security since RBAC limits what each user can access

Files should be Backed up to ensure that there is a disaster recovery set in place.

The Administrators should monitor the cloud to ensure that the health and performance is up to standard.

Azure Services that would be required

- Azure Virtual Machines

- Entra ID (Formerly Azure AD)
- Azure Back up
- Azure Blob storage
- Azure Monitor
- Azure SQL Database
- Azure App Service

Resource Groups

By making use of resource groups, you will be able to organize the cloud with greater effect for example Students that need to make use of LABS would be part of the Labs resource group and so on.

Region tags and Configuration

The region that would be used for CTU would be South Africa north and west since CTU is primary located within South Africa and these regions also covers all the Country and offers a connection to the cloud wherever you are located. I would create tags to help track where resources are being spent for example i would create a tag for each course to help track the cost and usage of the resources

Azure Service	Purpose	Week it is used in
Resource Group	This is a storage container that can hold any resources. These resources can also be managed as a group	Week 1
Virtual Network	Provides a fundamental building block that is required for an private network on the Azure platform. this servers allows VM's to be able to communicate with each other securely	Week 3
Virtual Machine	This is a computing resource that is scalable and is offered by the azure platform. VM's can also make use of different images such as Linux and Windows and more sub versions of these can also be chosen depending on the admin.	Week 3

Entra ID	This is an Identity management product that allows organization to implement different security protocols to the cloud one of these protocols is called zero trust.	Week 4
Network Security Group	This can be used to filter the incoming traffic via security rules that have been added.	Week 5
Azure Storage	This is a cloud storage solution that is used by azure. This storage has many key points such as High Availability and is very scalable.	Week 5
Microsoft Defender for cloud	This is a cloud protection tool that is used to ensure the safety of the cloud environment. It is used also to protect the applications on the cloud	Week 6
Azure monitor	This tool can be used to collect, analyze any telemetry from the cloud environment. You can use this to view the health of the cloud as well as performance of the environment.	Week 7
Azure cost management	This tool can be used to help monitor any of the outgoing costs of the cloud. It has features such as alerts that can be set up to alert an admin when the budget of a certain application is at certain points of the allocated budget.	Week 8

Screenshots

[Home](#) > Create a resource

Create a resource group ...

Basics

Tags

Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ

(Disabled) Azure for Students

Resource group name * ⓘ

RG_CTU

Region * ⓘ

(Africa) South Africa North

Basics

Tags

Review + create

Apply tags to your Azure resources to logically organize them by categories. A tag consists of a key (name) and a value. Tag names are case-insensitive and tag values are case-sensitive. [Learn more](#)

Name		Value	Resource	
Organisation	⌵	CTU	Resource group	🗑
IT Students	⌵	Education	Resource group	🗑
Owner	⌵	CTU_ADMIN	Resource group	🗑
Environment	⌵	CTU_LABS	Resource group	🗑
	⌵		Resource group	

Basics

Subscription	Azure for Students
Resource group name	RG_CTU
Region	South Africa North

Tags

Organisation	CTU
IT Students	Education
Owner	CTU_ADMIN
Environment	CTU_LABS

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